

A goalkeeper's performance is often hard to track and very often does not follow the team's performance, i.e., a goalie can play the best game of his/her life, but if the team doesn't score, the team can still lose the game. This can be demoralizing for the goalie if you are only tracking goals conceded. Data-driven statistic tracking can change the outlook from feelings about the game to objective takeaways that can be turned into training regimens and track improvements. No longer dealing with guesswork but patterns of prevention/interventions of the attack, transforming the goalie into a strategic asset rather than a purely reactive player.

Some important concepts to clarify:

Shot Situation describes the game context in which the shot occurred. Normal Game Play refers to a standard open-play shot, while Penalty Corner includes any first shot or rebound arising during a penalty-corner phase. Penalty Stroke refers to a formal penalty stroke, and 8-Second 1v1 refers to an eight-second, shootout-style attempt.

Shot Type describes the kind of shot taken within that situation. A 1st Shot is the first attempt in an attack, while a Rebound is a subsequent attempt following an initial save or deflection. In-Game 1v1 refers to a normal match-play one-on-one attempt rather than an eight-second shootout. Own Goal refers to a ball scored by a player's teammate into their own goal, while Other covers any shot that does not fit the listed categories.

Saves are recorded when the goalie directly prevents a shot or scoring attempt that would reasonably have resulted in a goal without the goalie's intervention. This can include stopping the ball with the pads, kickers, stick, gloves, body, or any controlled defensive action by the goalie. A Save should only be recorded when the attempt is genuinely on target or dangerous enough that the goalie's action prevents a goal. It should not be recorded when the attacker simply misses the goal, shoots wide, or loses control without the goalie causing the outcome. Saves are used in the Save Rate calculation and also form part of the Defence Rate calculation.

Angle Closed Off should be recorded when the goalie's movement, positioning, or presence removes the attacker's realistic scoring angle and forces a low-quality attempt, a missed shot, or no shot at all. It should not be used when the attacker simply misses an open chance through poor aim. This stat is important because it recognizes that a goalie's job is not only to save shots, but also to prevent dangerous scoring chances from developing. Angle Closed Off is included in Defence Rate, but not in Save Rate, because it is a successful defensive intervention rather than a direct save. In heat maps, Angle Closed Off is shown on the D Heat Map only, because there is no goal-box landing point to record.

- Angle Closed Off is included in **Defence Rate**.
- Angle Closed Off is not included in **Save Rate**.
- Angle Closed Off appears on the D heat map only, because there is no goal-box landing point.

Outnumbered Attempts are recorded when the goalie faces an attacking situation where the number of attackers is greater than the defensive cover available in the immediate scoring area. Examples include 2 vs 1, 3 vs 1, or 4+ vs 1 situations. Tracking this is important because goals conceded while outnumbered may not always reflect a technical failure by the goalie, but may point to a defensive structure problem in front of the goalie. A useful measurement is **Rate of Attempts at Goal Outnumbered % = Outnumbered Attempts at Goal ÷ Total Attempts at Goal × 100**. For this calculation, attempts at goal should include Saves and Goals Conceded, but not Angle Closed Off, because Angle Closed Off is a prevented scoring chance rather than a direct attempt on goal.

Consistent data entry is important if the stats are going to produce useful coaching insights. Every shot record should have one clear outcome: Save, Goal, or Angle Closed Off. Every shot should also have both a Shot Situation and a Shot Type, with Normal Game Play used as the default Shot Situation when no special situation applies. Rebound Result should only be used where relevant, especially after saves, and should identify whether the rebound went to a Safe or Dangerous area. Penalty Stroke attempts should always be recorded from the penalty spot on the D Heat Map. Angle Closed Off does not require a goal-box placement, because the ball did not create a normal goal-box save or goal location. Shootout data should be kept separate from normal match-play data so that reports remain accurate.

Data collected can be separated into two main categories: performance measuring and data to assist with training programs.

Performance Measuring Data:
5 stats that are measured in this category are save rate, Defence rate, rebound distribution rate, Expected Save Differential and Expected Defence Differential.

Save Rate:
Simply put, this is the percentage of shots on target faced (total scoring attempts that would have resulted in a goal without intervention) that were saved.
Save Rate % = (Saves ÷ Shots on Target) × 100
If you want to take it further, also differentiate between the first shot saved and saves/goals of the rebound to see if the goalie is recovering fast enough from the initial save.

Defence Rate
It can be said that the best save you can make is the one you don't have to! That is, often a goalie can close off the attacker's possible angle of attack, and the shot is taken missing the goals because the goalie had the goals covered. This save should be accredited to the goalie; this does not include shots where a goal was possible, but the attacker had bad aim.
Defence Rate % = **((Saves + Angles Closed Off) ÷ Total Shot Records) × 100**, where Total Shot Records = Saves + Goals Conceded + Angles Closed Off.

Rebound Distribution Rate
This arises from the thought that the goalie is not purely a defensive player but in fact can start the next offensive movement. For this, you record on every save whether the rebound went to a safe zone or a dangerous zone. A safe zone being either directly to an unmarked teammate or to an open area either side of the goals (not directly in the strike zone). Safe zone deflections + dangerous zone deflections should always = the total number of saves.
Rebound Distribution Rate % = (Safe Zone deflections ÷ Total Number of Saves) × 100

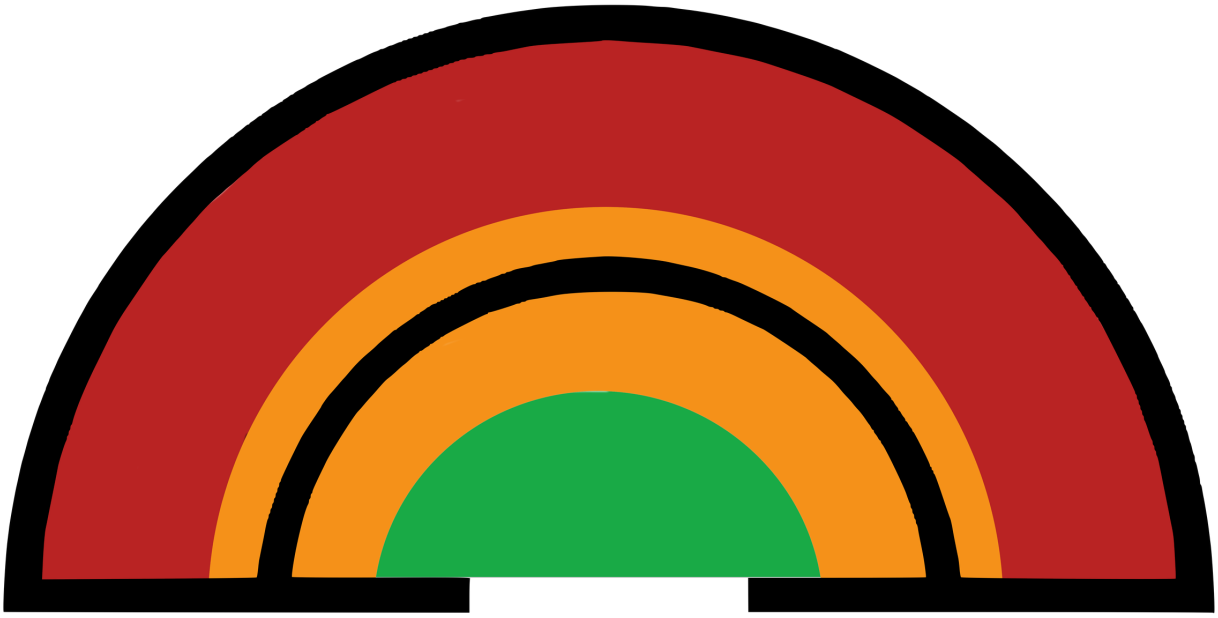
Shot Difficulty Weighting : Expected Save Differential and Expected Defence Differential
Basic statistics combined give a comprehensive picture of a goalkeeper's performance, but they share one fundamental flaw: every shot has the same value, whether it is hammered in from 1 m or rolled slowly from the top of the D. As a result, a goalkeeper who has only played against weaker teams has an inflated save rate compared with one who has played against stronger teams. A difficulty-weighted measure, the Expected Save Differential (xSD), addresses this.
In *Expanding Expected Goals Methodology in Field Hockey*, Myers and Daly (2022) modelled the likelihood of a shot becoming a goal using 1 012 shots from NCAA Division I field hockey. They found that the odds of a goal fall by about 8% for every additional yard between the shot and the centre of the goal. The odds rise by about 7% for every additional degree of shot angle, meaning the angle of goal mouth visible to the shooter, so central shots are more dangerous than shots from wide positions. For shots on target, a deflection increases the odds of a goal by about 119% (roughly 2,2 times), and a penalty stroke by about 700% (8 times), compared with a regular in-game shot.
A goalkeeper only has to deal with shots on target, so this framework uses the on-target figures as its base. Myers and Daly do not cover shot placement; the weights for this are the author's own estimates.
How the score works
Each shot is given an adjustment value between 0 and 1, found by multiplying together the adjustments for distance, angle, placement and any shot-specific factors below. The adjustment represents the goalkeeper's chance of stopping that shot. A value close to 1 is a routine shot; the lower the value, the harder the shot.

- A save scores 1 – adjustment

- A goal scores – adjustment

A save on a difficult shot therefore earns far more than a save on a routine shot, and a goal conceded from a routine shot costs far more than a goal from a difficult one. Every shot scores between −1 and +1, so no single shot can dominate the result.

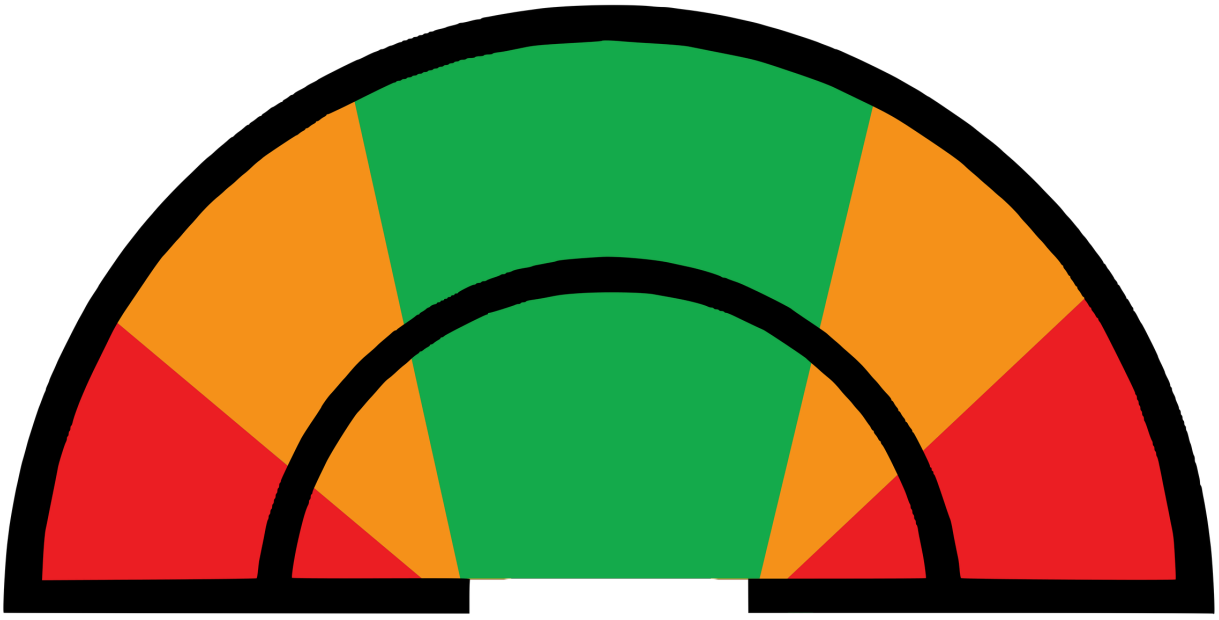
1: Adjust for distance



Green = 0,6 Orange = 0,88 Red = 0,97

Across the 16-yard circle, an 8% drop in odds per yard compounds to odds of about 0,28 at the edge of the circle compared with right at the goal, so closer shots are much harder to save. The weights are set so that an ordinary shot (middle distance, central, body height) has about a 70% chance of being saved, in line with typical save rates.

2: Adjust for angle



Green = 0,85 Orange = 0,93 Red = 0,98

Central shots have the widest view of the goal and are the hardest to save (green). Shots from near the baseline have the narrowest view and are the easiest (red).

3: Adjust for shot placement



Green = 0,65 Orange = 0,92 Red = 0,97

4: Goalie out numbered adjustments

In instances where the goalie is outnumbered, the chance of the shot being saved is greatly reduced. Outnumbered is recorded when there is no defender between the attackers and the goals and there are multiple attackers in this situation.

2v1 = 0,7

3v1 = 0,65

4+v1 = 0,6

5: Shot-specific adjustments

Tip-in / deflection = × 0,75. Myers and Daly found that a deflection makes a goal about 2,2 times more likely, measured in odds. For an ordinary shot with about a 70% chance of being saved, that lowers the chance of a save by about a quarter.

Penalty stroke = 0,25. Myers and Daly found that a penalty stroke makes a goal about 8 times more likely, measured in odds, which leaves the goalkeeper roughly a 25% chance of a save. This replaces the distance and angle adjustments; the placement adjustment still applies. For example, a stroke into a corner is 0,25 × 0,65 = 0,16, and a low stroke is 0,25 × 0,97 = 0,24.

Example: a deflected shot from the orange distance zone, green angle zone and green placement zone has an adjustment of 0,88 × 0,85 × 0,65 × 0,75 = 0,36. A save scores +0,64; a goal scores −0,36.

Rebound = × 0,80. Myers and Daly do not include rebounds, so this adjustment comes from ice hockey research. Macdonald, Lennon and Sturdivant (2012) found that a rebound, a shot taken within two seconds of a previous shot, has about 1,73 times the odds of becoming a goal compared with a first shot from the same distance and angle. The effect was larger at wider angles, and smaller when the shooter took their own rebound. This suggests the danger comes from a second attacker arriving while the goalkeeper is still out of position. For an ordinary shot with about a 70% chance of being saved, this lowers the chance of a save to about 57%, a multiplier of 0,82, rounded to 0,80. Because distance and angle were controlled in that study, the adjustment does not double count the fact that rebounds are usually close, central shots. No equivalent study exists for field hockey, so this weight should be checked against field hockey data as it is collected. How well the goalkeeper controlled the first save is measured separately by the Rebound Distribution Rate. Where a rebound is also an outnumbered attempt, both adjustments are applied, because the difficulties compound: the goalkeeper is out of position from the first save and also faces more attackers than defenders. This follows the expected goals models the weights are based on, where each factor raises the odds of a goal independently. Multiplying the adjustments is slightly cautious: compounding the original odds ratios would lower the chance of a save a little further.

In-Game 1v1 = × 0,70. No field hockey study has measured how much harder it is to save when an attacker is through on goal alone, so this adjustment comes from ice hockey breakaways.

Desjardins (2019) found that 25,6% of breakaway shots were goals, compared with 12,0% for ordinary shots from the same distance. In odds terms, that makes a goal about 2,5 times more likely, beyond what shot location explains. For an ordinary shot with about a 70% chance of being saved, this lowers the chance of a save to about 48%, a multiplier of 0,69, rounded to 0,70. In-Game 1v1 has its own value rather than using the outnumbered adjustment. A 1v1 is about the goalkeeper being isolated against a single attacker; outnumbered attempts involve two or more attackers and the option of a pass. Where an attempt is recorded as both, both adjustments are applied, because the extra attacker adds to the difficulty.

Penalty Corner = × 1,00 (no adjustment). Penalty corner shots are often hit or flicked hard from a set position with a screen in front, so it would be reasonable to expect them to be harder to save. However, Myers and Daly (2022) tested penalty corners as a variable in both of their models, and it was not significant in either once distance, angle and shot type were included. In their on-target model, which the weights here are based on, flicks were not significant either. Distance, angle, placement and the deflection adjustment therefore already account for the difficulty of corner shots. No study has measured the effect of screening in field hockey. International penalty corner conversion is about 21% per corner (Blanford, 2024), but that counts every corner, including those that never produce a shot on target, so it is not a measure of how hard corner shots are to save.

8 Second 1v1 / 6 Second Shoot-out = × 0,60. A shoot-out works like a penalty stroke: one attacker, no defenders and a set start. Unlike a stroke, the shot can be taken from anywhere in the D, so this is a multiplier on the distance, angle and placement adjustments rather than a fixed value. In ice hockey, Desjardins (2019) found that 32,7% of shoot-out attempts were goals, about 1,4 times the odds of a breakaway. That is about 3,5 times the odds of an ordinary shot from the same distance. For an ordinary shot with about a 70% chance of being saved, this lowers the chance of a save to about 40%, a multiplier of 0,60. Field hockey data supports this: Houta (2020) analysed 119 elite shoot-out attempts and found 56,3% were goals, with misses included. With this multiplier, a typical shoot-out shot from close to middle range has an adjustment of about 0,25 to 0,36. That predicts 64% to 75% of shots on target becoming goals, which is consistent with Houta's figure once missed attempts are allowed for. No indoor data exists, so the same value is used for the indoor 6 Second Shoot-out.

Outnumbered: 2 vs 1 = × 0,70, 3 vs 1 = × 0,65, 4+ vs 1 = × 0,60. No field hockey study has measured how being outnumbered affects the chance of a goal, so this adjustment comes from ice hockey. Desjardins (2019) found that 20,4% of shots on 2-on-1 rushes were goals, compared with 8,3% for ordinary shots from the same distance, about 2,5 to 2,8 times the odds. NHL tracking data shows that an extra attacking player almost triples the chance of a goal (Lukan, 2021). For an ordinary shot with about a 70% chance of being saved, odds 2,5 times higher lower the chance of a save to about 48%, a multiplier of about 0,70. No research covers 3 vs 1 or 4+ vs 1, so these values are the author's estimates, with each extra attacker adding a little less because the goalkeeper can only be beaten once. Shot location is already counted by the distance and angle adjustments, and the ice hockey figure was measured against shots from the same distance, so it does not double count. Applying this adjustment reflects the point made earlier that goals conceded while outnumbered often point to a problem in the defensive structure rather than the goalkeeper.

Minimum adjustment = 0,05. Because adjustments compound, a few rare combinations, such as a deflected rebound in a 3 vs 1 from close range into a corner, produce values close to zero. At that point a goal costs the goalkeeper almost nothing and a save earns almost a full point, from one attempt. To stop a single attempt from having too much influence, no adjustment is allowed to fall below 0,05, which still leaves the goalkeeper only a 5% chance of a save.

6: Calculating the Expected Save Differential (xSD)

Add the scores of every shot faced and divide by the total number of shots:

xSD = (Σ saves (1 − adjustment) − Σ goals (adjustment)) ÷ total shots

The adjustments added together give the number of saves expected for the shots faced, so xSD is the number of saves made above or below that expectation, per shot. An xSD of 0 means the goalkeeper performed as expected for the shots faced. A positive xSD means they saved more than expected; a negative xSD means fewer.

Worked example:

- Goalkeeper A faces 10 routine shots (long range, wide angle, low: 0,97 × 0,98 × 0,97 = 0,92), making 8 saves and conceding 2 goals, an 80% save rate.
  xSD = (8 × 0,08 − 2 × 0,92) ÷ 10 = **−0,12** (−12%)

- Goalkeeper B faces 10 difficult shots (close range, central, into a corner: 0,60 × 0,85 × 0,65 = 0,33), making 5 saves and conceding 5 goals, a 50% save rate.
  xSD = (5 × 0,67 − 5 × 0,33) ÷ 10 = **+0,17** (+17%)

Despite the higher save rate, Goalkeeper A performed below expectation: shots that routine should almost always be saved, so conceding two of them costs a lot. Goalkeeper B performed well above expectation relative to the difficulty of the shots faced. xSD follows the same principle as the "Goals Saved Above Expected" measure used in ice hockey and football analytics, and gives a fairer way to compare goalkeepers who have faced different levels of opposition.

Expected Defence Differential (xDD)

The same method can be applied to Defence Rate to give the Expected Defence Differential (xDD). Angles Closed Off are included and scored like a save (1 − adjustment). Because the ball does not reach the goal, they use only the distance and angle adjustments, not placement. Closing down an attacker who is close and central therefore earns much more than closing down one who is wide and far out.

xDD = (Σ saves (1 − adjustment) + Σ angles closed off (1 − adjustment) − Σ goals (adjustment)) ÷ (total shots + angles closed off)

It is important to note that xSD and xDD are not as widely accepted as Save Rate %, but they are still useful metrics to record and compare.

Other interesting statistics to record if possible include:
Goals & saves outnumbered, i.e., multiple attacks against the goalie
Goals & saves from penalty corners
Goals & saves from penalty strokes
Goals & saves from 8-second 1v1
Goals & saves from goalie challenging attacker 1v1 in game
Own goals scored by teammates

Suggested Method of Data Collection

There are two main ways of recording one is to record video and capture stats later from the video this is first prize if it can be done correctly as it also gives footage that can be used in training and the stats can be recorded out of the heat of the game and shots can be reviewed until the correct stat is recorded, or you can record the stats manually during the game either on paper or in a dedicated app.

If recording stats manually during a game, a shorthand for data collection is suggested to keep up with the fast-paced game. Recording save data first, then graphical data used for training second. This shorthand is a data entry for each attempt at goals. Then later, after the match, use the data collected to collate stats for the game.

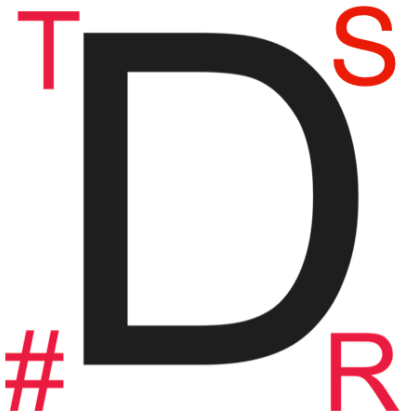
D: Main Data Point
- S = Save
- G = Goal
- A = Angle Closed Off

S: Shot Situation
- "" = Normal Game Play
- PC = Penalty Corner
- PS = Penalty Stroke
- V = 8 Second 1v1

T: Type of Shot
- "" = 1st Shot at Goal
- R = off Rebound
- C = Challenged Attacker 1v1
- O = Own Goal scored by teammate
- D = Tip in or Deflection

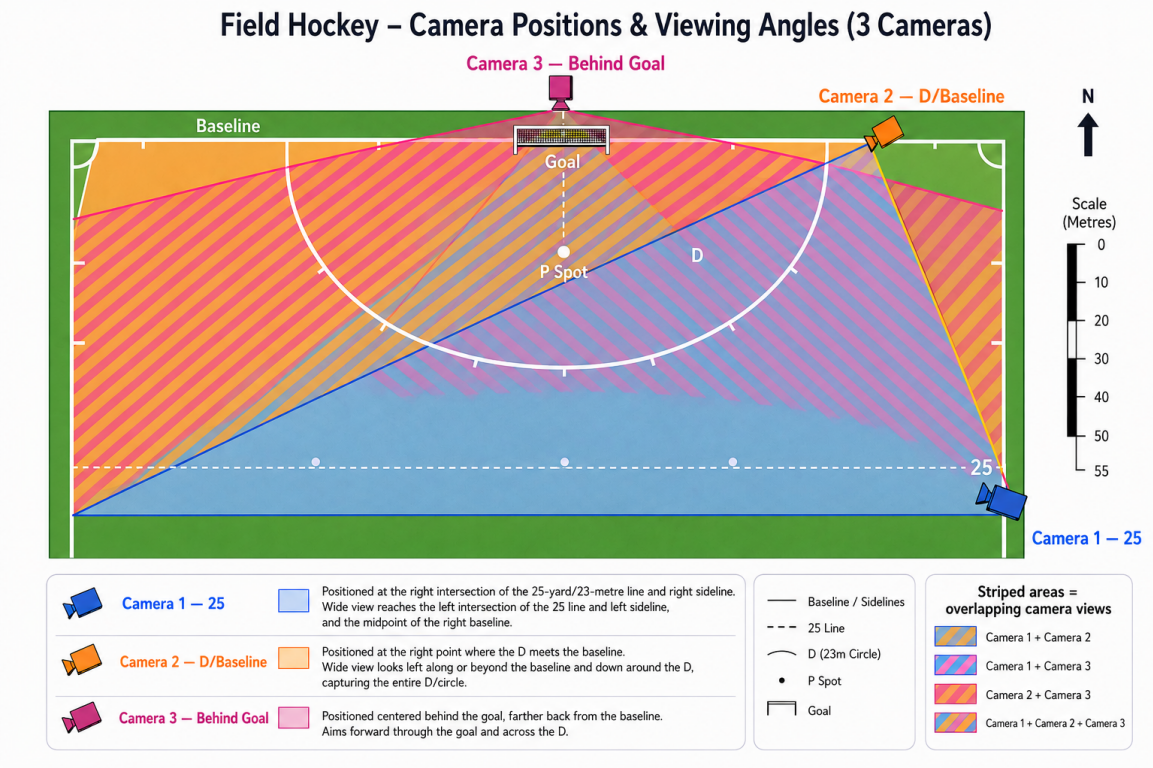
#: Number of Attackers VS Goalie in D

R: Rebound Zone
- S = Safe
- D = Dangerous



Filming the game find the angle that works best for you a higher tripod is suggested 3-5m to look above people heads looking down onto the turf here are some suggestions but angle may vary

based on where you can set up and where is available on the turf and multiple angles is always advantageous as it may give you a better view of some shots hidden by players or umpires.



Performance Benchmarking and Analysis
Raw data only results in coaching insights when interpreted through a framework of benchmarks. Suggested performance benchmarks.

Level	Benchmark Save Rate
U13	65–75%
U16	70–80%
1st Team	75–85%
Provincial	80%

If possible, don't view one Match as a standalone; rather, look at a 5-match trend.

The Interpretative Framework:
High Saves + High Goals: This indicates a systemic failure in the team's defensive structure, allowing an excessive volume of high-quality entries.

Low Save %: Typically points to individual technical lapses, such as late reactions, poor positioning, or a failure to be "set" before the shot.

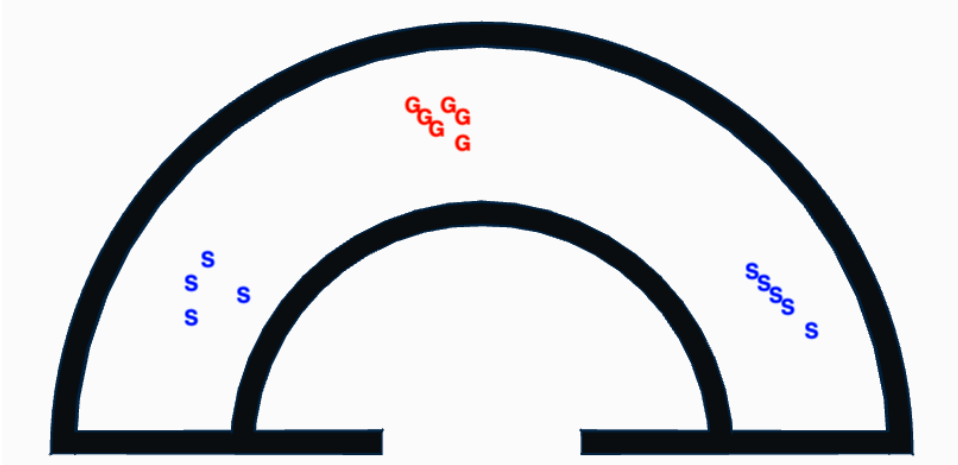
Frequent Rebound Goals: Indicates a need for a concentrated focus on kicking width and clearing to the sidelines.

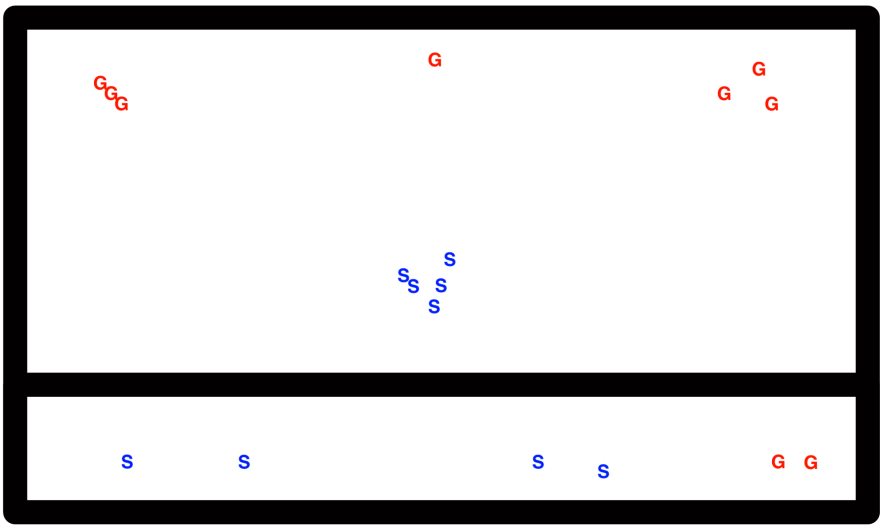
Qualitative Command and Organization
Subjective coaching ratings provide context to the numbers. Evaluating the goalie as the "defensive organizer" whose vocal presence reduces the overall shots faced.

The 1–5 Performance Rating Scale:
Defensive Organization: Effectiveness in directing markers and managing the structural integrity of the back line.
Penalty Corner Communication: Clarity and timing of instructions during the set-piece setup and execution.
Overall Presence: The ability to remain loud, confident, and physically imposing to the opposition.
Consultant Intervention: A score of 1 or 2 requires an immediate tactical intervention. The goalkeeper should call out marking assignments at least 5 seconds before circle entry to ensure the defensive line remains efficient and proactive.

Data to assist with training programs
This requires a different type of data to be recorded. This is the How, not the What.
Data points to record are **Where was the shot taken from** Record from which area in the D was the shot taken from and whether it was a goal or save. This will show a pattern of which areas of the D the goalie struggles to save shots from, and coaches should train shots from this area. This can be recorded visually by marking a diagram of the D and creating a heat map.

While one match's data can show a specific team's shooting style/abilities, after a few matches, a trend may be visible, and a training program can be developed to work on areas that goals are often scored from. **Where the Shot lands** Similarly, by marking where in the goal box the goal is scored, a training program can be developed to work on weaker areas.





Shootout / 8 Second 1v1 Summary Stats
Shootout attempts and 8 Second 1v1 situations should be analysed separately from normal match-play shots because the tactical situation is different. In a shootout, the goalie faces a structured 1v1 from a set starting point, whereas normal match-play shots develop from open play, penalty corners, rebounds, or defensive breakdowns. For this reason, shootout data should be shown as a summary rather than mixed directly into normal shot totals. Useful shootout stats include Shootout Rounds, Shootout Saves, Shootout Goals Conceded, Shootout Save Rate, Shootout Defence Rate, and Most Common Save Method. This allows the coach to evaluate the goalie's 1v1 performance without distorting normal match statistics.

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